Benchmark: **MA.7.AR.4.1** · Is it proportional? (table, graph, description) Student code: _____ Calculator: yes

Learning goal: I can decide whether a relationship is proportional. (circle your level after scoring)

4	I can explain why a relationship is not proportional using more than one representation.
3	I can decide whether a table, graph, or written description shows a proportional relationship and say why.
2	I can check whether two ratios are equal.
1	I need help to start this benchmark.

1. Select **all** the situations that describe a proportional relationship.

Select **all** the correct answers.

A. ☐ A printer prints 30 pages every minute.

B. ☐ A phone plan costs $36 plus $0.10 for each text message.

C. ☐ A candle is 12 inches tall and burns down 1 inch each hour (its height compared with the hours burned).

D. ☐ A taxi charges $3.00 plus $2.00 per mile.

E. ☐ A store sells oranges at $2.40 per pound.

2. Does the table show a proportional relationship between x and y?

x	y
1	5
2	10
4	23
5	25

A. Yes, because $y \div x$ is the same for every pair of values.

B. No, because the x-values do not increase by the same amount.

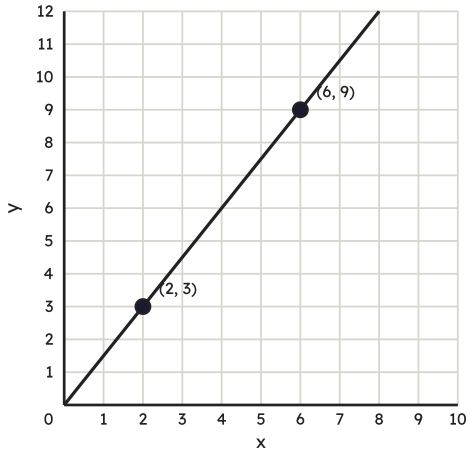
C. No, because $y \div x$ is not the same for every pair of values.

D. Yes, because the first two pairs of values have the same ratio.

Turn over →

Benchmark: **MA.7.AR.4.1** · Is it proportional? (table, graph, description) Student code: _____

3. Does the graph show a proportional relationship between x and y?



A. Yes, because the line is straight and passes through the origin.

B. Yes, because y increases whenever x increases.

C. No, because the line does not reach the edge of the grid.

D. No, because the line does not pass through the origin.

4. A machine takes 15 minutes plus 10 minutes for each box packed.

Is the relationship between the boxes packed and the total minutes proportional?

A. Yes, because there is no setup time at the start, only 10 minutes each time.

B. Yes, because the minutes added for each box packed never change.

C. No, because there is a setup time of 15 minutes before the first box packed.

D. No, because the minutes and the boxes packed are measured in different units.

5. The table shows values of x and y.

x	y
1.5	1.8
2	2.4
2.5	3
5	6

Noor says the relationship cannot be proportional because the table does not include $x = 0$.

Is Noor's reason correct? Is the relationship proportional? If it is, give the constant of proportionality; if it is not, name a pair whose ratio is different.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.4.1 · MA.7.AR.4.1 · Is it proportional? (table, graph, description)

Unit 11: Representing Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. A printer prints 30 pages every minute; E. A store sells oranges at $2.40 per pound.**

A relationship is proportional only when the two quantities have a constant ratio and both start at 0.
Proportional: "A store sells oranges at $2.40 per pound." — the same price for every pound, and 0 pounds cost $0; "A printer prints 30 pages every minute." — the same pages every minute, and 0 minutes print 0 pages.
Not proportional: "A phone plan costs $36 plus $0.10 for each text message." — the $36 is charged even with 0 texts; "A candle is 12 inches tall and burns down 1 inch each hour (its height compared with the hours burned)." — the candle is 12 inches tall at hour 0; "A taxi charges $3.00 plus $2.00 per mile." — the $3.00 charge is paid before the first mile.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

2. Regular **C. No, because $y \div x$ is not the same for every pair of values.**

Check $y \div x$ for each pair: $5 \div 1 = 5$, $10 \div 2 = 5$, $23 \div 4 = 5.75$, $25 \div 5 = 5$.
The ratios are not all the same, so the relationship is not proportional.

3. Regular **A. Yes, because the line is straight and passes through the origin.**

A graph shows a proportional relationship when it is a straight line through the origin $(0, 0)$.
This line passes through $(0, 0)$, so it is proportional.

4. Regular **C. No, because there is a setup time of 15 minutes before the first box packed.**

A proportional relationship must pass through $(0, 0)$ and have a constant ratio.
The starting setup time of 15 means 0 boxes packed already gives 15 minutes, so it is not proportional.

5. Rigor **Noor's reason is not correct; the relationship is proportional, with constant of proportionality 1.2. Score: 1 point for all three parts (the reason judged, the proportional decision, and the constant or a pair with a different ratio, in any equivalent form); two parts earn 0.**

A table does not have to show $(0, 0)$; test the ratio $y \div x$ for every pair.
$1.8 \div 1.5 = 1.2$, $2.4 \div 2 = 1.2$, $3 \div 2.5 = 1.2$, $6 \div 5 = 1.2$.
Every ratio is 1.2, so the relationship is proportional and the constant of proportionality is 1.2.

Re-check strip (3 items, C-AR.4.1-R)

1. Regular **B. A store sells grapes at $1.85 per pound; C. A recipe uses 7 cups of flour for every 4 loaves of bread.**

A relationship is proportional only when the two quantities have a constant ratio and both start at 0.
Proportional: "A store sells grapes at $1.85 per pound." — the same price for every pound, and 0 pounds cost $0; "A recipe uses 7 cups of flour for every 4 loaves of bread." — the same flour for every loaf, and 0 loaves need 0 cups.
Not proportional: "A gym charges $38 to join plus $11 each month." — the $38 joining fee is paid before the first month; "A phone plan costs $34 plus $0.15 for each text message." — the $34 is charged even with 0 texts; "A candle is 9 inches tall and burns down 1 inch each hour (its height compared with the hours burned)." — the candle is 9 inches tall at hour 0.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

2. Regular **D. Yes, because $y \div x$ is the same for every pair of values.**

Check $y \div x$ for each pair: $3 \div 1 = 3$, $9 \div 3 = 3$, $12 \div 4 = 3$, $15 \div 5 = 3$.
Every ratio is the same, so the relationship is proportional.

3. Regular **C. No, because there is a charge of 15 dollars before the first pound of apples.**

A proportional relationship must pass through $(0, 0)$ and have a constant ratio.
The starting charge of 15 means 0 pounds of apples already gives 15 dollars, so it is not proportional.

Comparison strip (2 items, C-AR.4.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **A. An empty tank fills with 4 liters of water each minute; C. A worker earns $22 for each hour worked; E. A store sells oranges at $1.60 per pound.**

A relationship is proportional only when the two quantities have a constant ratio and both start at 0.
Proportional: "A store sells oranges at $1.60 per pound." — the same price for every pound, and 0 pounds cost $0; "A worker earns $22 for each hour worked." — the same pay every hour, and 0 hours earn $0; "An empty tank fills with 4 liters of water each minute." — the same liters every minute, and the tank starts empty.
Not proportional: "A taxi charges $2.00 plus $3.25 per mile." — the $2.00 charge is paid before the first mile; "A plant is 7 inches tall and grows 1 inch each week (its height compared with the number of weeks)." — the plant is already 7 inches tall at week 0.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

2. Regular **86**

Every other pair gives $y \div x = 11$.
For $x = 8$, a proportional relationship needs $y = 88$, not 86.

Reteach plan — C-AR.4.1

MA.7.AR.4.1 · Is it proportional? (table, graph, description)

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.4.1-R

Benchmark: **MA.7.AR.4.1** · Is it proportional? (table, graph, description) Student code: _____ Calculator: yes

1. Select **all** the situations that describe a proportional relationship.

Select **all** the correct answers.

A. ☐ A gym charges $38 to join plus $11 each month.

B. ☐ A store sells grapes at $1.85 per pound.

C. ☐ A recipe uses 7 cups of flour for every 4 loaves of bread.

D. ☐ A candle is 9 inches tall and burns down 1 inch each hour (its height compared with the hours burned).

E. ☐ A phone plan costs $34 plus $0.15 for each text message.

2. Does the table show a proportional relationship between x and y?

x	y
1	3
3	9
4	12
5	15

A. Yes, because y increases whenever x increases.

B. No, because $y \div x$ is not the same for every pair of values.

C. No, because the x-values do not increase by the same amount.

D. Yes, because $y \div x$ is the same for every pair of values.

3. A stand charges 15 dollars plus 12 dollars for each pound of apples.

Is the relationship between the pounds of apples and the total dollars proportional?

A. Yes, because the dollars added for each pound of apples never change.

B. No, because the dollars and the pounds of apples are measured in different units.

C. No, because there is a charge of 15 dollars before the first pound of apples.

D. Yes, because there is no charge at the start, only 12 dollars each time.

Score: ____ / 3 → copy to your progress strip

Benchmark: **MA.7.AR.4.1** · Is it proportional? (table, graph, description) Student code: _____ Calculator: yes

1. Select **all** the situations that describe a proportional relationship.

Select **all** the correct answers.

A. ☐ An empty tank fills with 4 liters of water each minute.

B. ☐ A plant is 7 inches tall and grows 1 inch each week (its height compared with the number of weeks).

C. ☐ A worker earns $22 for each hour worked.

D. ☐ A taxi charges $2.00 plus $3.25 per mile.

E. ☐ A store sells oranges at $1.60 per pound.

2. The table does **not** show a proportional relationship.

x	y
2	22
4	44
6	66
8	86

Which value of y would have to change for the relationship to be proportional?

Score: ____ / 2 → **copy to your progress strip**